

Module 9: Linear Programming — Summary

What this module is about. Linear programming turns a real decision problem into a mathematical model. The variables describe the choices we control. Constraints describe what is allowed. The objective function describes what we want to make as large or as small as possible. Convex geometry from Module 8 supplies the method: find the feasible vertices, evaluate the objective at them, and compare.

A two-variable linear program contains:

1. **Decision variables:** quantities being chosen, such as acres of two crops.
2. **Constraints:** linear inequalities representing limited resources or minimum requirements.
3. **Objective:** a linear expression such as profit or cost.
4. **Direction:** maximize or minimize the objective.

Unless negative quantities make sense in context, include $x \geq 0$ and $y \geq 0$.

Example 1: Translate a farming problem

A farmer plants corn (x acres) and soybeans (y acres). Corn uses 3 labor-hours per acre and earns \$200 profit per acre. Soybeans use 2 labor-hours per acre and earn \$150 profit per acre. At most 120 labor-hours and 50 acres of land are available.

The model is

$$\begin{aligned} 3x + 2y &\leq 120 && \text{(labor),} \\ x + y &\leq 50 && \text{(land),} \\ x &\geq 0, \quad y &\geq 0, \end{aligned}$$

with objective

$$\text{maximize } P = 200x + 150y.$$

Words such as *at most* produce $\leq$ constraints. Minimum requirements often produce $\geq$ constraints.

1. Write the variables, constraints, objective, and direction.
2. Determine the feasible region.
3. Find all feasible vertices.
4. Evaluate the objective at every vertex.
5. Choose the largest value for maximization or smallest value for minimization.
6. State the answer in the original context.

Example 2: Maximization at the vertices

Maximize

$$f = 5x + 8y$$

subject to

$$x + 2y \leq 16, \quad 3x + 2y \leq 24, \quad x, y \geq 0.$$

The feasible vertices are

$$(0, 0), (8, 0), (4, 6), (0, 8).$$

Evaluate:

$$0, \quad 40, \quad 68, \quad 64.$$

The maximum is 68 at $(4, 6)$.

Example 3: Minimization with minimum requirements

Minimize $C = 3x + 5y$ subject to

$$x + y \geq 8, \quad 2x + y \geq 10, \quad x, y \geq 0.$$

The feasible region extends away from the origin because the requirements say the variables must be large enough. Relevant boundary vertices are $(8, 0)$, $(2, 6)$, $(0, 10)$. Their costs are

$$24, \quad 36, \quad 50.$$

The minimum is 24 at $(8, 0)$.

Important modeling point. A continuous linear program may produce a fractional solution. If the variables must represent whole objects or whole servings, the continuous optimum is a guide; nearby feasible integer points must then be checked separately.

Module 9: Linear Programming — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Workflow

Define variables → write constraints → write objective and direction → find feasible vertices → evaluate objective → interpret the optimum.

1. A farmer plants corn (x acres) and soybeans (y acres). Corn requires 3 labor-hours per acre and earns \$200 profit per acre. Soybeans require 2 labor-hours per acre and earn \$150 profit per acre. The farmer has at most 120 labor-hours and 50 acres of land.
 - (a) Write the labor constraint.
 - (b) Write the land constraint.
 - (c) Write the nonnegativity constraints.
 - (d) Write the profit objective and state whether it should be maximized or minimized.

2. Maximize

$$f = 5x + 8y$$

subject to

$$x + 2y \leq 16, \quad 3x + 2y \leq 24, \quad x, y \geq 0.$$

The feasible vertices are $(0, 0)$, $(8, 0)$, $(4, 6)$, $(0, 8)$. Evaluate f at all four vertices and identify the optimal point.

3. A cost problem has minimum requirements:

$$x + y \geq 8, \quad 2x + y \geq 10, \quad x, y \geq 0,$$

and objective $C = 3x + 5y$. The relevant boundary vertices are $(8, 0)$, $(2, 6)$, $(0, 10)$. Evaluate the cost at each point and identify the minimum.

4. A feasible region has vertices

$$(0, 0), (4, 0), (3, 2), (0, 5).$$

Maximize $f = 3x + 3y$.

- (a) Evaluate f at all four vertices.
 - (b) Which vertices are optimal?
 - (c) What can you conclude about the edge joining the two optimal vertices?
5. A nutrition problem produces a continuous optimum of $x = 2.4$ servings and $y = 1.7$ servings. The cafeteria can serve only whole servings. Explain what should be done next before choosing a practical solution.

Module 9: Linear Programming — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

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1. A caterer makes sandwiches (x) and salads (y). Each sandwich uses 2 oz of bread and 3 oz of filling. Each salad uses 1 oz of bread and 4 oz of filling. The caterer has 80 oz of bread and 150 oz of filling. Sandwiches earn \$4 and salads earn \$5. Write the complete linear program, including objective direction.
 2. A feasible region has vertices $(0, 0)$, $(6, 0)$, $(4, 5)$, $(0, 7)$. Maximize $R = 6x + 7y$ by evaluating all four vertices.
 3. A least-cost problem has objective $C = 5x + 2y$ and feasible boundary vertices $(6, 0)$, $(3, 6)$, $(0, 12)$. Evaluate the cost and identify the minimum.
 4. A linear objective takes the same maximum value at two adjacent vertices of a feasible polygon. What is true about the points on the edge joining those vertices?
 5. A factory's mathematical optimum is $x = 7.5$ machines and $y = 3$ machines. If only whole machines can be purchased, why is $(7.5, 3)$ not yet a practical final answer, and what kind of points should be checked next?